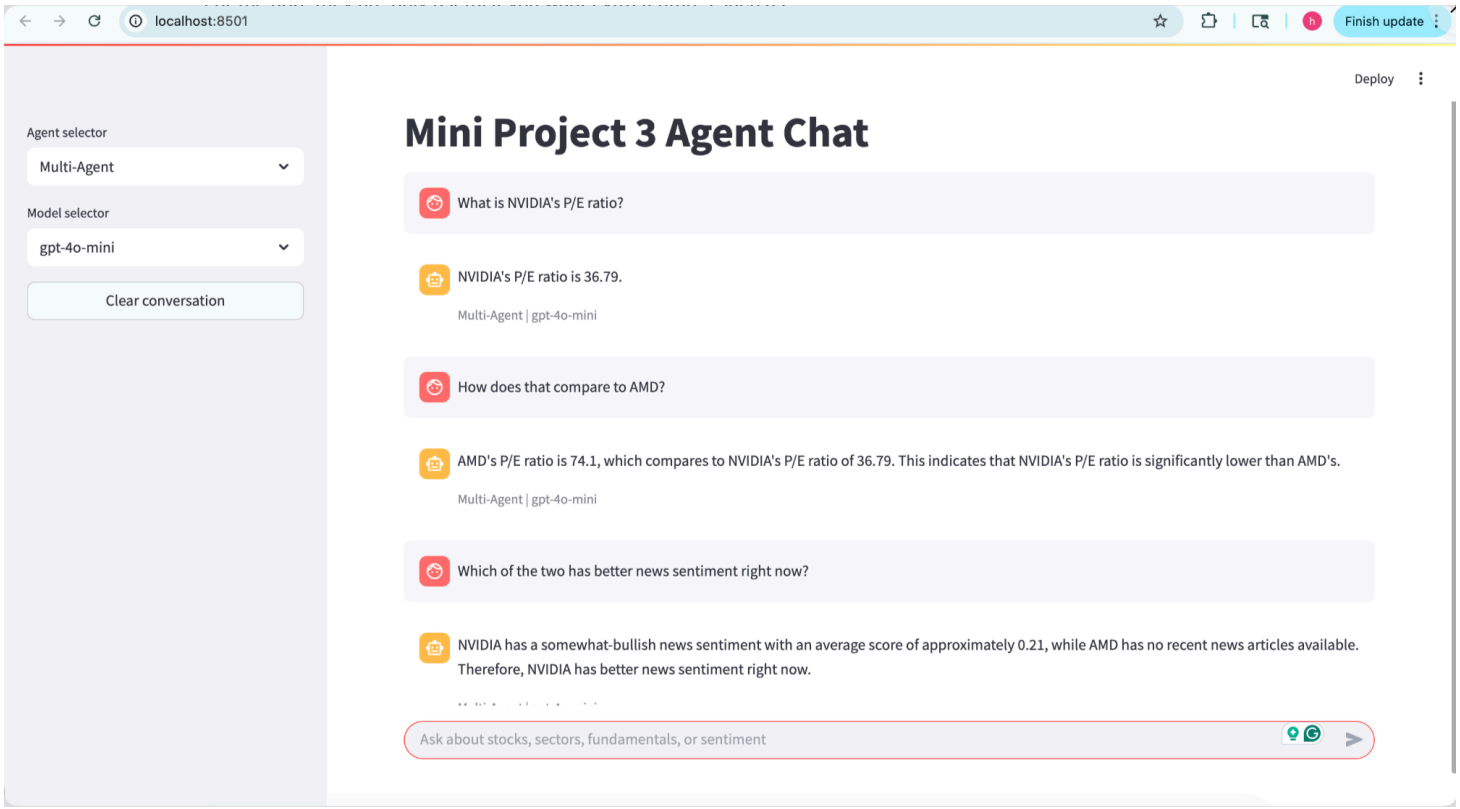
Mini Project 3: Multi-Agent Financial Chat — Screenshot Documentation

This document presents screenshots from the deployed Streamlit application, demonstrating both agent architectures (Single Agent and Multi-Agent) with two model backends (gpt-4o-mini and gpt-4o). Each screenshot shows a complete 3-turn conversation where follow-up questions are correctly resolved using conversational context.

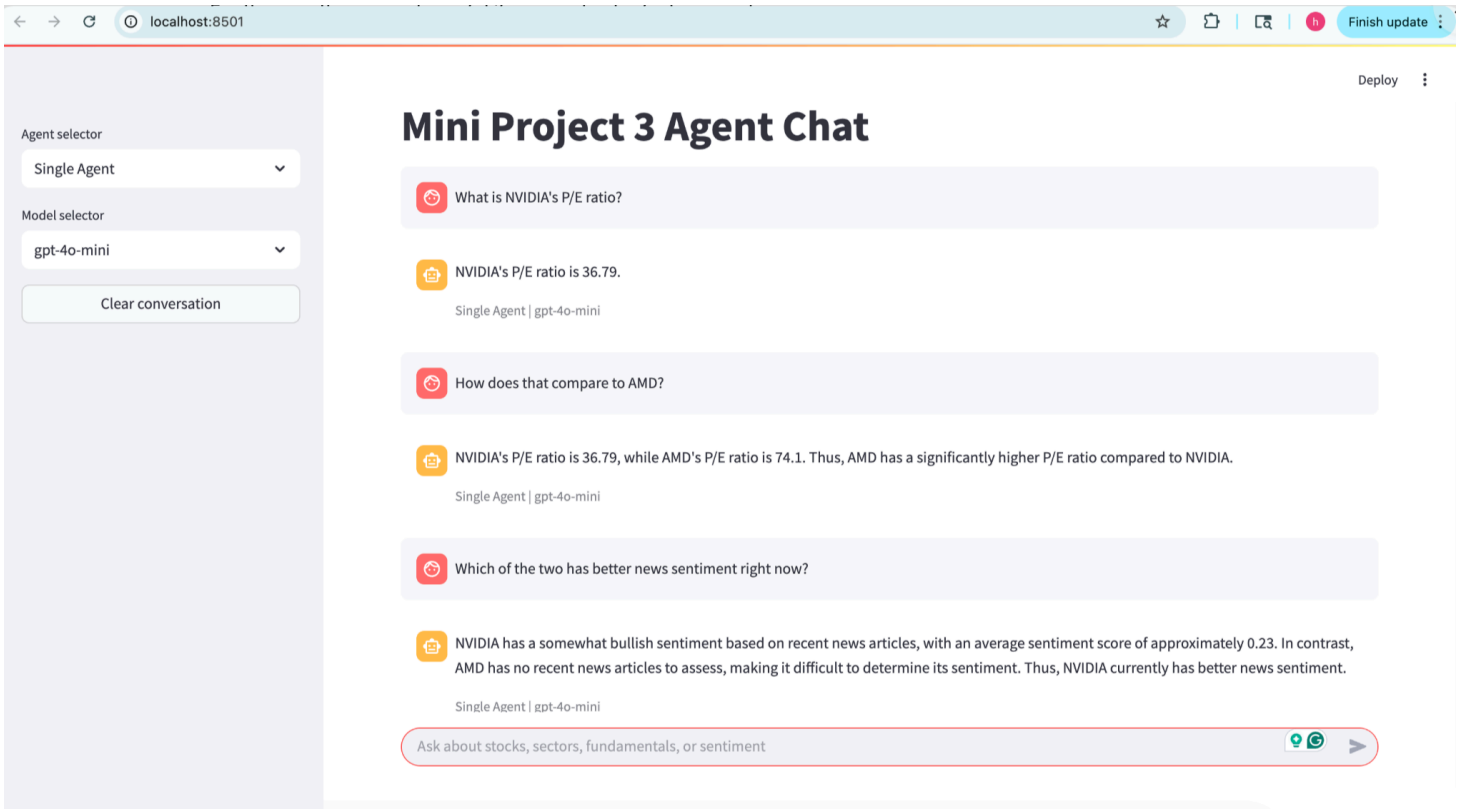
Screenshot 1: Multi-Agent | gpt-4o-mini

Three-turn conversation (NVIDIA vs AMD): P/E ratio lookup, comparison, and news sentiment analysis.



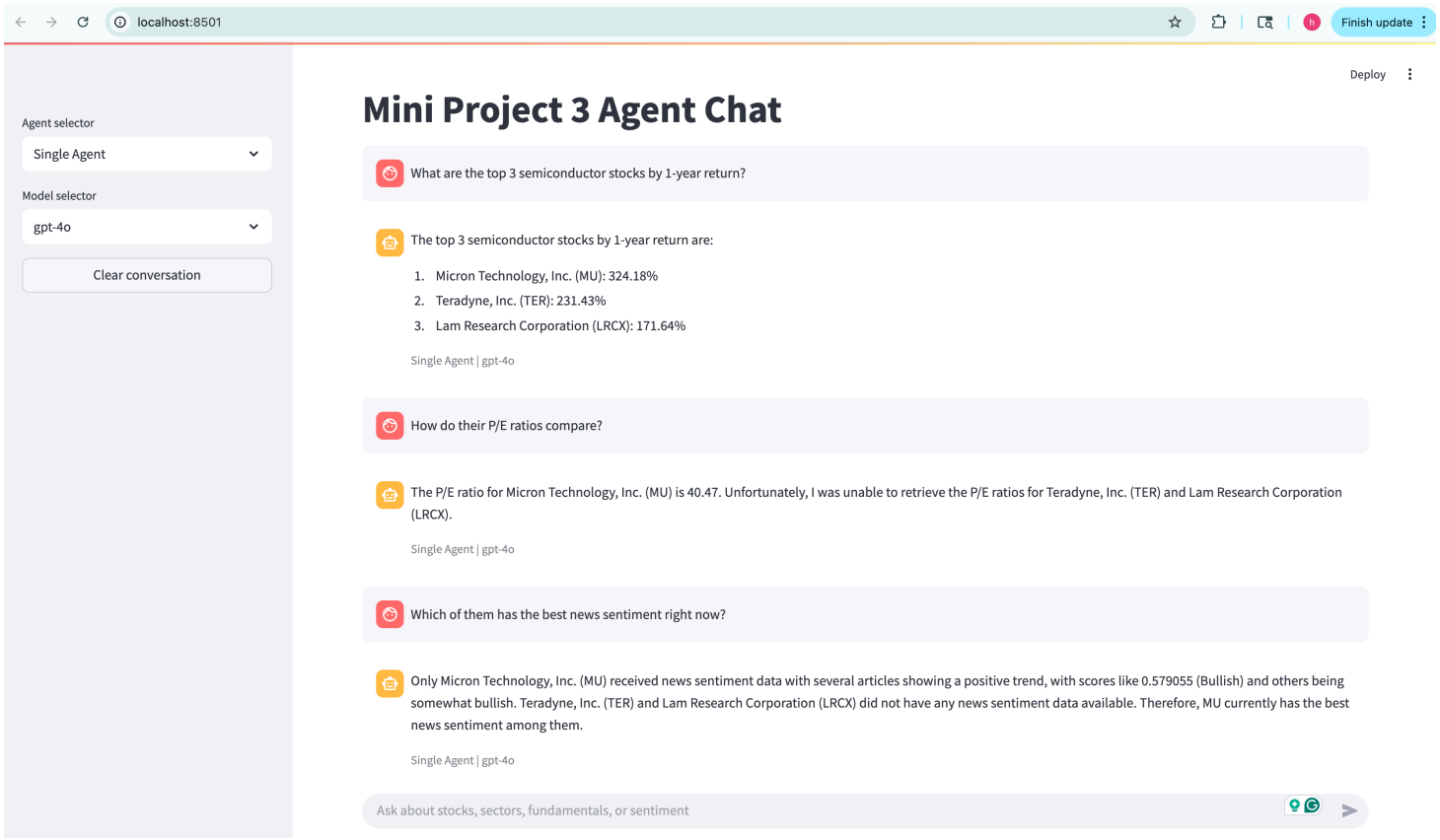
Screenshot 2: Single Agent | gpt-4o-mini

Same three-turn conversation using the single-agent architecture.



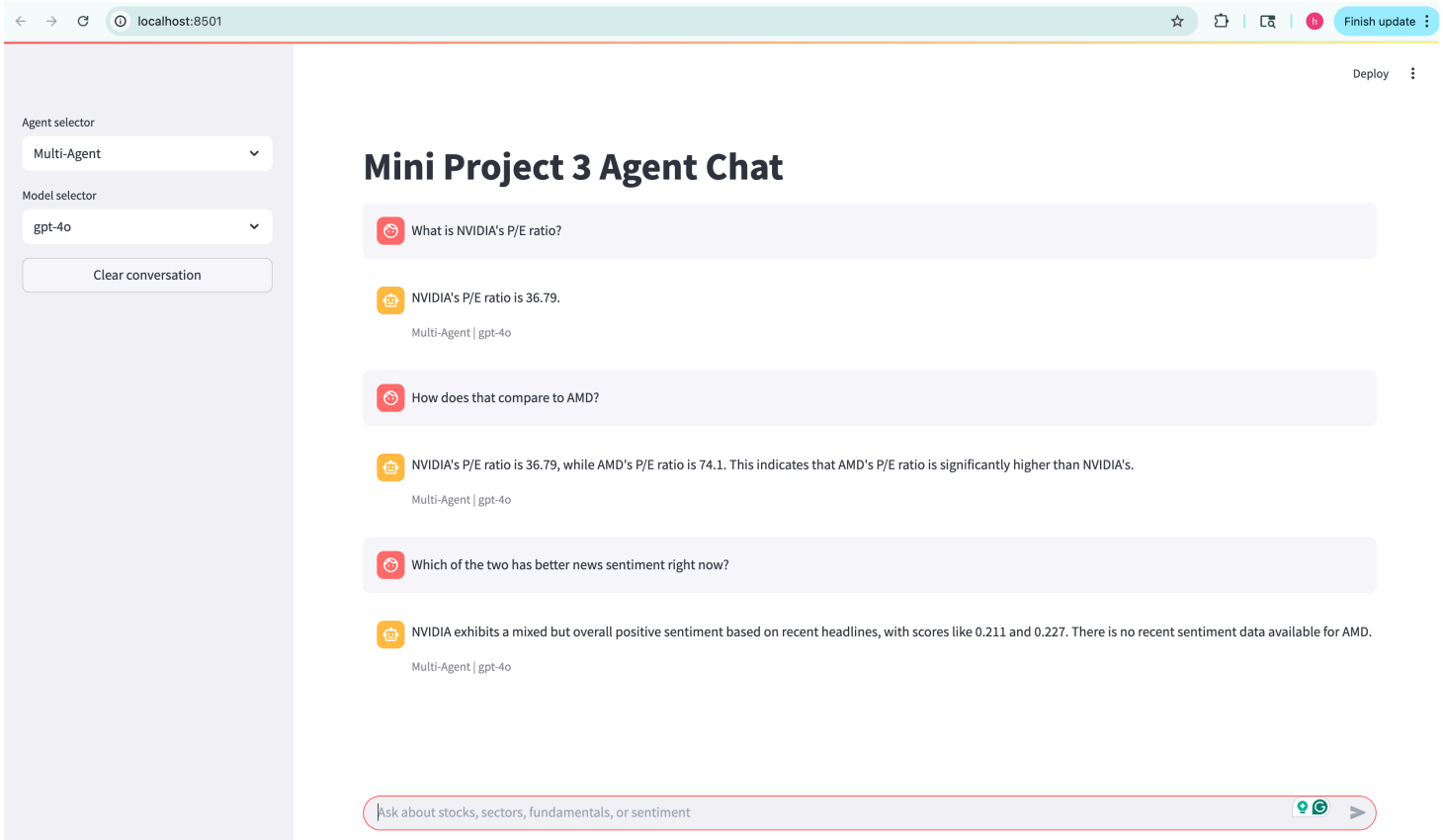
Screenshot 3: Single Agent | gpt-4o

Three-turn conversation (NVIDIA vs AMD) with gpt-4o.



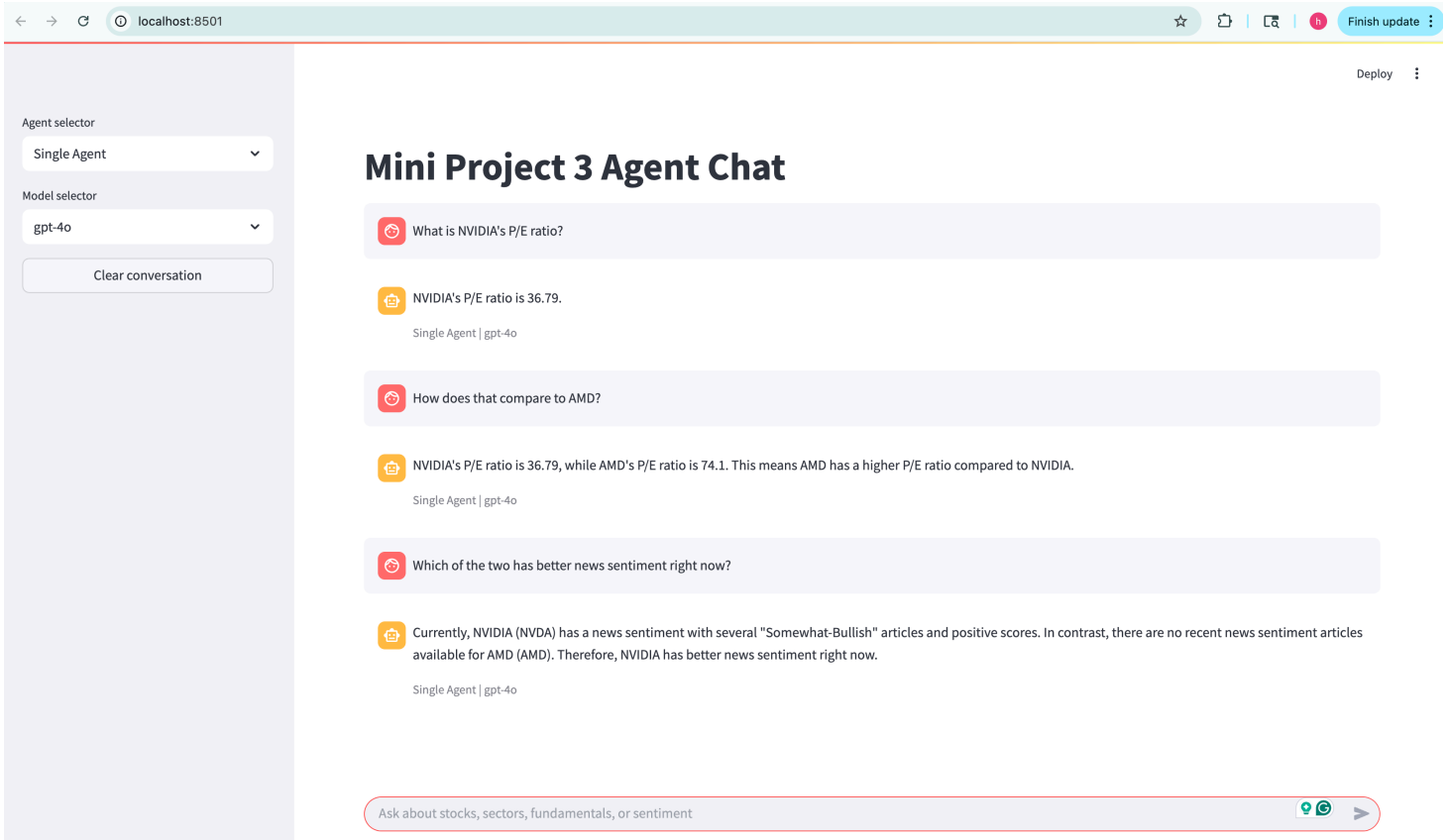
Screenshot 4: Multi-Agent | gpt-4o

Same NVIDIA vs AMD conversation using the multi-agent architecture with gpt-4o.



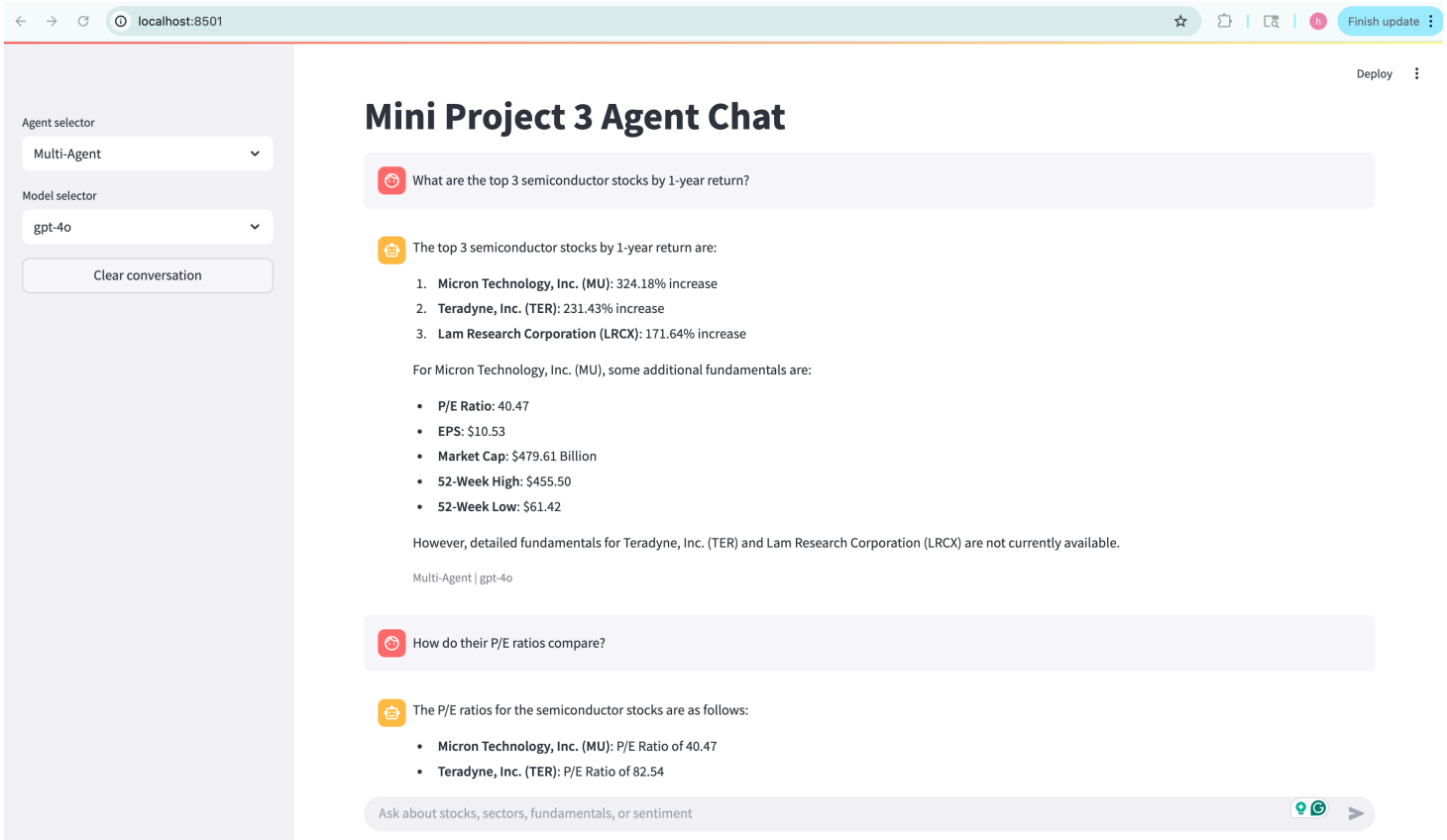
Screenshot 5: Single Agent | gpt-4o (Semiconductor Sector)

Three-turn conversation: top semiconductor stocks by 1-year return, P/E comparison, and best news sentiment.



Screenshot 6: Multi-Agent | gpt-4o (Semiconductor Sector, Page 1)

Multi-agent version of the semiconductor query with additional fundamentals retrieved.



Screenshot 7: Multi-Agent | gpt-4o (Semiconductor Sector, Page 2)

Continuation: P/E ratio comparison and news sentiment results.

localhost:8501

Finish update

Deploy

Agent selector

Multi-Agent

Model selector

gpt-4o

Clear conversation

- **P/E Ratio**: 40.47
- **EPS**: $10.53
- **Market Cap**: $479.61 Billion
- **52-Week High**: $455.50
- **52-Week Low**: $61.42

However, detailed fundamentals for Teradyne, Inc. (TER) and Lam Research Corporation (LRCX) are not currently available.

Multi-Agent | gpt-4o

How do their P/E ratios compare?

The P/E ratios for the semiconductor stocks are as follows:

- **Micron Technology, Inc. (MU)**: P/E Ratio of 40.47
- **Teradyne, Inc. (TER)**: P/E Ratio of 82.54

The P/E Ratio for Lam Research Corporation (LRCX) is currently unavailable.

Multi-Agent | gpt-4o

Which of them has the best news sentiment right now?

Currently, Micron Technology, Inc. (MU) has the best news sentiment, with a generally positive and bullish outlook. There is no recent news sentiment data available for Teradyne, Inc. (TER) or Lam Research Corporation (LRCX).

Multi-Agent | gpt-4o

Ask about stocks, sectors, fundamentals, or sentiment